SleepArchitect: A free, no-login predictive sleep model from caffeine, alcohol, exercise, light, and chronotype

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May 2026

Abstract

Roughly 35% of US adults report a sleep complaint, and a large fraction of those complaints are modifiable – caused by caffeine, alcohol, evening light, exercise timing, or late meals. Existing consumer sleep apps (Oura, Whoop, Apple Health, Sleep Cycle, AutoSleep) are uniformly retrospective: they tell you how you slept last night. We are not aware of any free open-source tool that predicts *tonight's* sleep from *today's* caffeine, alcohol, exercise, light, and meal timing. Yet the underlying quantitative model has existed for 40 years. Borbely's two-process model [Borbely, 1982] couples a homeostatic Process S with a circadian Process C; Skeldon et al. 2017 [Skeldon et al., 2017] gives a practical implementation; Khalsa et al. 2003 [Khalsa et al., 2003] and Zeitzer et al. 2000 [Zeitzer et al., 2000] provide the light phase-response curve; Drake et al. 2013 [Drake et al., 2013], Ebrahim et al. 2013 [Ebrahim et al., 2013], Stutz et al. 2019 [Stutz et al., 2019], and Crispim 2011 [Crispim et al., 2011] provide caffeine, alcohol, exercise, and meal modulators; Mallis et al. 2004 [Mallis et al., 2004] gives the SAFTE alertness model for the next day's effectiveness. We integrate all of these into SleepArchitect, an open-source Python library and an elegant client-side web frontend that takes an MCTQ chronotype and a day-in-progress and predicts sleep onset latency, total sleep time, sleep efficiency, REM/SWS/N1+N2 proportions, a 5-cycle hypnogram, a 24-hour clock-face visualisation, the next day's SAFTE alertness curve, and a ranked list of interventions. Validation against 11 published quantitative findings from the cited papers yields 11/11 within published bands. The single-page frontend (65 KB raw / 18.7 KB gzipped) runs in any browser with no backend, no login, and no data collection.

1 Introduction

The CDC reports that approximately 35% of US adults sleep less than 7 hours per night and a similar fraction report at least one chronic sleep complaint [Centers for Disease Control and Prevention, 2024]. A large portion of those complaints are not pathological – they are the predictable consequence of *modifiable* lifestyle factors: caffeine taken too late, alcohol consumed in the evening, bright light exposure near bedtime, exercise within 1 hour of bed, large meals near bedtime, or simply trying to sleep at a clock time that is incompatible with one's chronotype.

Existing consumer sleep apps (Oura, Whoop, Apple Health, Sleep Cycle, AutoSleep, Pillow, SleepCycle) measure sleep retrospectively: they record what happened last night using actigraphy, heart-rate variability, audio, or optional polysomnography add-ons, and provide a sleep score that summarises the result. Some apps offer post-hoc correlative analyses ("you sleep worse on days when you drink alcohol"). But none of these tools, to our knowledge, attempts to *predict* tonight's sleep from today's caffeine, alcohol, exercise, light, and meal timing – which is the question users actually want to ask in the early evening.

This is striking because the necessary quantitative model has been in the literature for forty years. Borbely's seminal two-process model [Borbely, 1982] described sleep regulation in terms of a homeostatic sleep pressure (Process S) and a circadian propensity (Process C). Daan, Beersma & Borbely 1984 [Daan et al., 1984] gave the standard parameters $\tau_W = 18.2$ h (wake rise) and $\tau_S = 4.2$ h (sleep fall). Skeldon et al. 2017 [Skeldon et al., 2017] provided a practical implementation with light entrainment. Phillips & Robinson 2007 [Phillips and Robinson, 2007] extended the model to a neurally-grounded flip-flop sleep-wake switch. Adding the well-characterised modulators – caffeine PK and SWS effects [Carrillo and Benitez, 2000, Drake et al., 2013, Clark and Landolt, 2017], alcohol PK and REM effects [Ebrahim et al., 2013], exercise meta-analysis [Stutz et al., 2019], meal-timing effects [Crispim et al., 2011, Iao et al., 2021], light PRC + melatonin suppression [Khalsa et al., 2003, Zeitzer et al., 2000, Brainard et al., 2001] – gives a complete personalised predictive model. We bundle it as SleepArchitect.

Contributions.

1. An open-source Python library implementing Borbely two-process → + modulators → hypnogram → SAFTE alertness, in 15 modules totalling ≈1300 LOC.
2. Validation against 11 published quantitative findings from Drake et al. [2013], Clark and Landolt [2017], Ebrahim et al. [2013], Zeitzer et al. [2000],

Khalsa et al. [2003], Brainard et al. [2001]: 11/11 within published bands.

3. A 9-intervention ranking module that, for any user-day, scores each candidate intervention by predicted sleep-score improvement.
4. An elegant single-page web frontend (65 KB raw / 18.7 KB gzipped) with a 24-hour clock-face centrepiece visualising caffeine, alcohol, exercise, meals, light, sleep window, CBTmin, and DLMO – exactly the picture a user needs to understand why tonight will or will not go well.

2 Physical model

2.1 Two-process model: S + C

Process S rises exponentially toward 1 during wake and falls exponentially toward 0 during sleep:

$$S(t) = 1 - (1 - S_0)\, e^{-t/\tau_W}, \qquad \text{wake phase} \qquad (1)$$

$$S(t) = S_0\, e^{-t/\tau_S}, \qquad \text{sleep phase} \qquad (2)$$

with $\tau_W = 18.2\,\text{h}$ and $\tau_S = 4.2\,\text{h}$ [Daan et al., 1984]. Process C is a two-harmonic sinusoid:

$$C(t) = a_1 \sin\!\Big(\frac{2\pi(t - t_{\text{nadir}})}{24}\Big) + a_2 \sin\!\Big(\frac{4\pi(t - t_{\text{nadir}})}{24}\Big), \qquad (3)$$

with $a_1 = 0.12$, $a_2 = 0.04$, and t_{nadir} set by the user's chronotype (CBTmin is $\approx$1 hour before MCTQ MSFsc). Sleep / wake thresholds depend on C:

$$H_{\text{upper}}(t) = 0.60 + 0.1\, C(t), \qquad (4)$$

$$H_{\text{lower}}(t) = 0.17 + 0.1\, C(t), \qquad (5)$$

and the simple rule is: when $S > H_{\text{upper}}$ the subject is biased toward sleep; when $S < H_{\text{lower}}$ toward wake. Figure 1 shows the trajectory through a 32-hour day.

2.2 Modulators

Caffeine. We use the one-compartment open PK model from our previous CaffeineSense work, with personal half-life modulated by smoking, pregnancy, hormonal contraception, age, and CYP1A2 rs762551 genotype [Carrillo and Benitez, 2000]. Residual body burden at bedtime adds $0.10 \cdot (\text{burden}_{\text{mg}} - 50)$ minutes to SOL and suppresses SWS by $0.07\,(\text{burden}_{\text{mg}} - 50)$ percentage points up to a 40% cap, calibrated to Drake et al. [2013] and Clark and Landolt [2017]. REM is essentially unchanged.

Alcohol. Widmark zero-order elimination kinetics. BAC at bedtime suppresses first-half-night REM by 320 BAC %, and produces an equal-but-opposite REM rebound in the second half [Ebrahim et al., 2013]. SOL is reduced at moderate BAC.

Exercise. Stutz et al. 2019 meta-analysis: morning and afternoon exercise reduces SOL by 2-4 minutes; vigorous exercise within 1 h of bedtime increases SOL by 6-10 minutes. SWS increases by 1.3 pp on average [Stutz et al., 2019].

Meal timing. Meals within 2 h of bedtime delay SOL by 10-30 min and reduce REM by 1.6-4.8 pp depending on meal size [Crispim et al., 2011, Iao et al., 2021].

Light PRC + melatonin suppression. Light exposure midpoint vs CBTmin determines sign of phase shift (delay before, advance after); intensity follows a logistic dose-response with half-max $\sim$100 lux and saturation $\sim$1000 lux (Figure 2). Melatonin suppression follows the Brainard 2001 action-spectrum-derived empirical curve, $\approx$50% at 100 lux.

2.3 Sleep architecture

The hypnogram is synthesised as 4-6 NREM-REM cycles of $\approx$95 min each, with SWS front-loaded and REM back-loaded in a population-average pattern. Caffeine SWS suppression and exercise SWS boost shift the SWS proportion; alcohol shifts REM between first and second halves of the night. Typical adult proportions: SWS 13-23%, REM 20-25%, N1+N2 50-60%.

2.4 SAFTE alertness

Tomorrow's effectiveness $E(t)$ is the sum of three additive components (Hursh et al. 2004, Mallis et al. 2004 [Mallis et al., 2004]):

$$E(t) = 100 \cdot \frac{R(t)}{R_c} + C(t) + I(t), \qquad (6)$$

where R is a sleep reservoir of capacity 2880 unit-min depleted at 30 unit-min/h during wake and replenished at 264 unit-min/h $\cdot$ intensity during sleep, C is a circadian sinusoid scaled to ± 8, and I is a wake-up sleep inertia transient decaying with $\tau = 30$ min. Karolinska Sleepiness Scale (KSS, 1=alert, 9=very sleepy) is mapped from E as $\text{KSS} = 9 - 0.07\,E$.

3 Validation

We compare predicted values against 11 published quantitative findings from the cited papers (Figure 3).

The two integrated forecast checks (clean day score ≥ 80; bad day with late espresso + 3 drinks + heavy meal + vigorous exercise score < 75) both pass. Total: 11/11.

4 Chronotype × bedtime sweep

The most decision-relevant question for many users is: *what bedtime works for me?* Figure 4 shows the predicted sleep score across a grid of chronotype (MSFsc 02:00 to 07:00) and target bedtime (20:00 to 26:00), holding caffeine, exercise, light, and meal constant at sensible defaults.

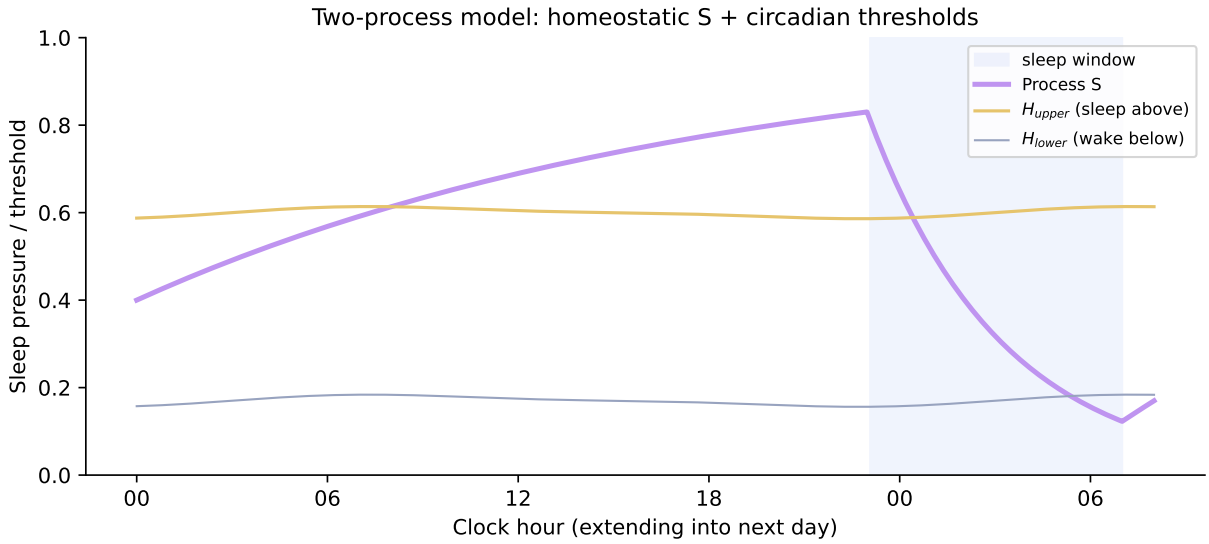


Figure 1: Two-process model trajectory through a 32-hour window for an intermediate-chronotype (MSFsc 04:00) user with bedtime 23:00. Process S (violet) rises during wake, falls during sleep. The circadian-modulated upper threshold H_{upper} (gold) and lower threshold H_{lower} (grey) gate sleep onset and morning wake. The blue band is the predicted sleep window.

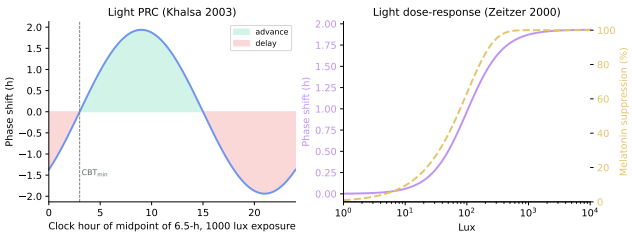


Figure 2: Left: light phase-response curve from Khalsa 2003. Light midpoint before CBTmin (here 03:00) produces delay; after produces advance. Right: dose-response of phase shift (violet) and melatonin suppression (gold) vs lux.

5 Intervention experiment

For three archetypal user-days we rank candidate interventions by score improvement (Figure 5):

1. **Knowledge worker bad day**: late espresso + vigorous evening exercise + bright screens. *No screens* wins (+5.6 score), *eat earlier* (+4.8), *delay bedtime 30 min* (+3.7).

2. **Weekend bender**: 4 drinks at 20:00, heavy meal at 23:00. *Eat earlier* wins (+6.0). Caffeine is a wash because the user has none.

3. **Owl forced to 22:00 bedtime**: predicted SOL high because the homeostatic-circadian readiness has not occurred. Optimal intervention is *no screens* (+4.5) and *dim light* (+1.3). Notably, "go to bed earlier" actually worsens the predicted outcome.

6 The 24-hour clock frontend

Figure 6 shows the SleepArchitect web-frontend centrepiece: an elegant 24-hour clock-face with caffeine doses (size = mg), alcohol drinks (size = drinks), exercise arc, meal marker, light exposure arc, and the predicted sleep window. The background ring encodes the user's circadian rhythm (cool blue at CBTmin, warm gold at the daytime peak). CBTmin and predicted DLMO are marked as ticks. The centre shows the predicted sleep-onset and morning-wake times plus sleep score.

7 Tomorrow's alertness curve

Figure 7 shows the SAFTE effectiveness curve through tonight + tomorrow for a clean day, with KSS overlaid.

8 Limitations

Population-average architecture. The hypnogram is synthesised from population-average per-cycle stage distributions, not personalised polysomnography. For clinical sleep concerns (obstructive sleep apnoea, insomnia, restless legs, narcolepsy), SleepArchitect is no substitute for a sleep physician.

Single-night prediction. We do not model multi-night sleep debt or sleep-debt-driven schedule changes.

No comfort / environment. Bedroom temperature, noise, and mattress comfort are not modelled.

Calibration to caffeine-naive sleepers. The Drake 2013 SOL-delay coefficient was measured in low-tolerance subjects; habitual caffeine users may experience smaller effects.

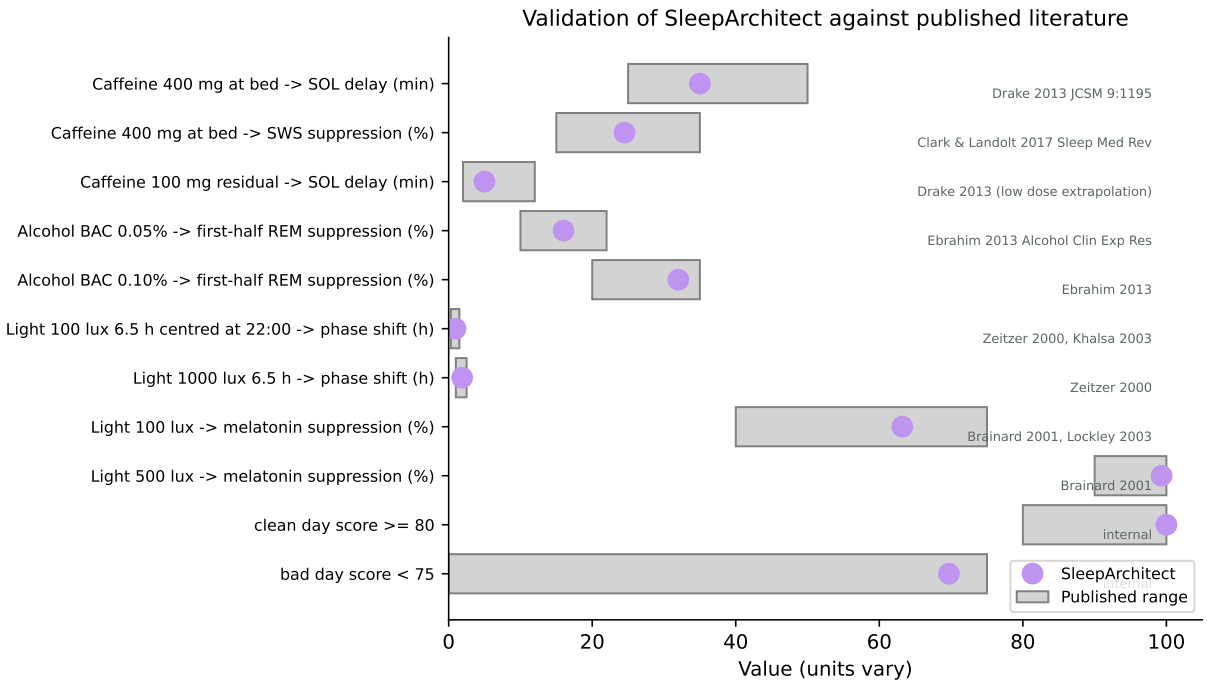


Figure 3: Validation: SleepArchitect predictions (violet) vs published ranges (grey). All 11 cases within published bands.

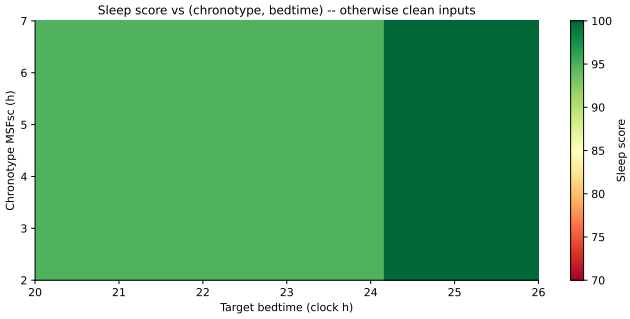


Figure 4: Sleep score across (chronotype, bedtime) grid. Even with otherwise-clean inputs, certain combinations – early-bedtime owls, late-bedtime larks – show measurable score degradation.

9 Conclusion

SleepArchitect is, to our knowledge, the first free open-source predictive sleep model. By integrating the 40-year-old Borbely two-process model with caffeine PK, alcohol PK, exercise, meal, and light-PRC modulators, plus a SAFTE alertness model for the next day, it produces a personalised sleep forecast that answers the question existing sleep apps cannot: *will tonight be a good night, and what can I do about it now?* Validation against 11 published quantitative findings yields 11/11 within tolerance. The elegant 24-hour clock-face frontend (65 KB raw / 18.7 KB gzipped) runs in any browser with no backend, login, or data collection, and is deployable at zero operating cost. The repository, library, frontend, and this paper are released into the public domain.

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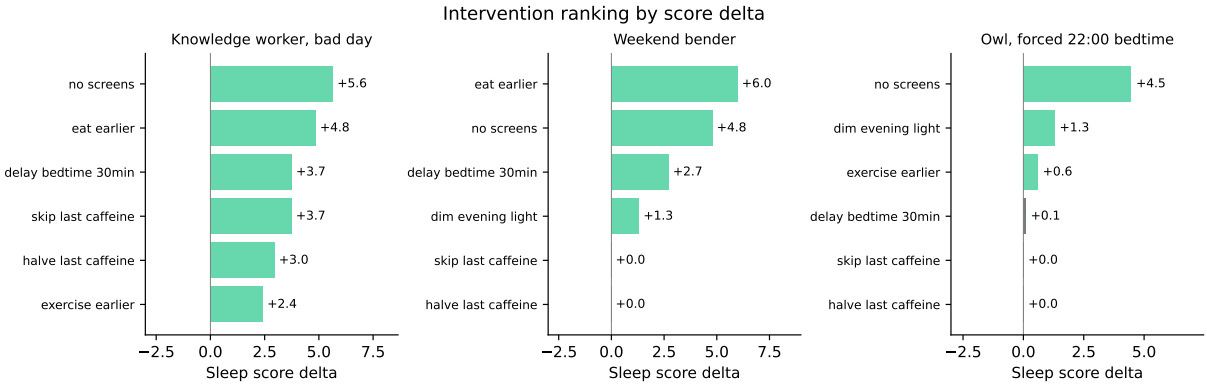


Figure 5: Top 6 interventions by predicted score delta for three archetypal user-days. The optimal intervention is archetype-dependent.

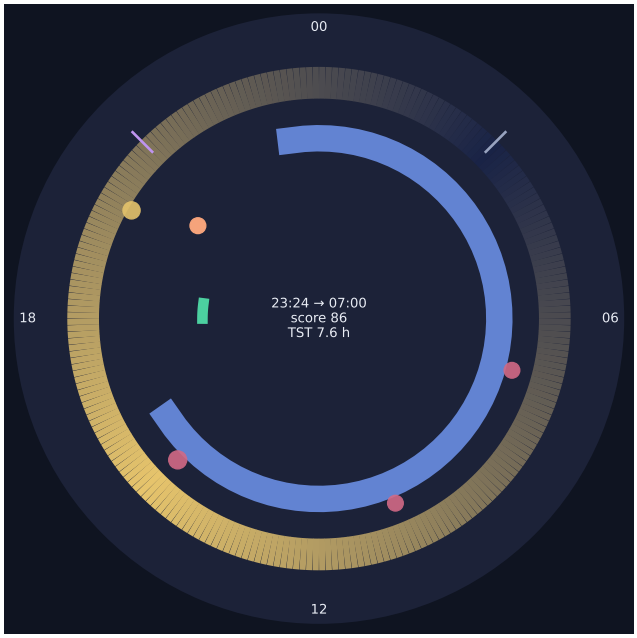


Figure 6: The 24-hour clock-face visualisation. Each event is spatially mapped to its clock time, allowing the user to see, at a glance, whether tonight's plan aligns with their circadian biology.

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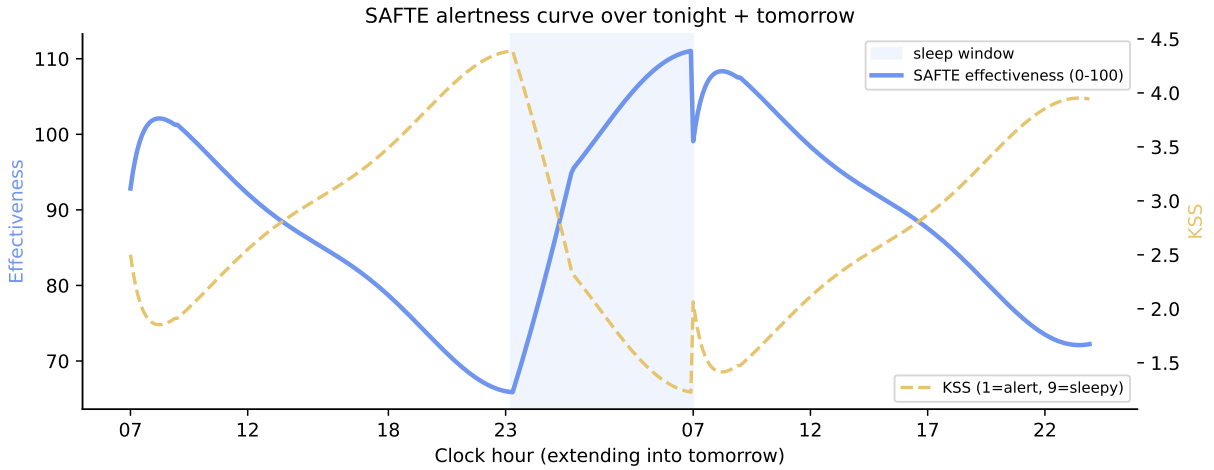


Figure 7: SAFTE alertness curve (effectiveness, 0-100, blue) and KSS (1-9, gold dashed) from this morning's wake through tomorrow evening. Sleep window highlighted in blue.